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Bridging Artificial Intelligence and Real Intelligence: Self-Scaffolding Computational Modeling with Generative AI in Chemistry

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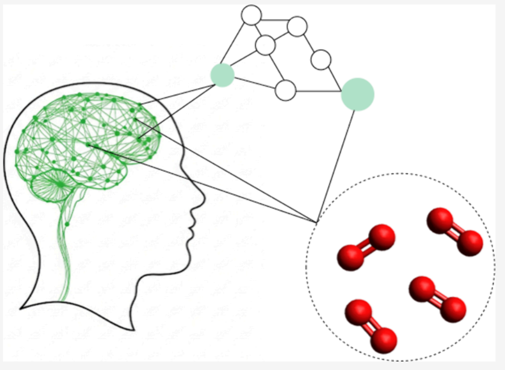
ABSTRACT: Chemistry education researchers have, for many years, explored different ways of learning chemistry through modeling and open-ended problem-solving. With the emergence of generative artificial intelligence (GenAI) tools such as ChatGPT, students now have access to dynamic scaffolds that can potentially support them in modeling. However, we still know little about how these tools function within such contexts. This study investigates how students in higher education use GenAI to engage in complex computational modeling in chemistry. Using cognitive clinical interviews with think-aloud protocols, the role of GenAI in students' modeling activities is analyzed through the lens of distributed cognition. The findings from this study reveal four distinct patterns of GenAI use: (1) Leveraging AI to retrieve and clarify information, (2) Using GenAI to test and critique ideas, (3) Outsourcing thinking to the AI, and (4) GenAI output exceeds student understanding. Productive interactions occurred when students actively orchestrated the AI as part of a distributed cognitive system, using it to retrieve, evaluate, and build upon their own reasoning. In contrast, unproductive use occurred when students outsourced thinking to the AI or were overwhelmed by outputs they could not integrate into their existing knowledge. These findings suggest that effective use of GenAI in chemistry education depends not only on technical proficiency, but also on students' ability to structure prompts, evaluate AI output, and retain control of the problem-solving process. We argue that the use of GenAI should be explicitly addressed in chemistry instruction and propose that students be taught to engage with GenAI as part of a distributed cognitive system— − retaining executive control, providing appropriate context, and iteratively refining their inquiries— − to support meaningful engagement and learning in chemistry.



KEYWORDS: *Generative Artificial Intelligence, LLMs, ChatGPT, Higher Education, Modeling, Computational Chemistry*

■ INTRODUCTION

Chemistry, with its intricate web of concepts, symbols, and applications, can be a demanding domain that requires students to blend analytical problem-solving skills with conceptual knowledge, simulations, and practical skills in the laboratory or with fieldwork.[1−3] In order to learn and use these skills, students have access to many different tools, both functional (e.g., calculators and software) and theoretical (e.g., textbooks). Within the past couple of years, a new tool has emerged: chatbots based on Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs).

Many educators and educational researchers hold significant concerns, but also great optimism about how these generative AI tools will impact chemistry education and practice:[4] given their newness, we do not know if using GenAI will harm or improve student learning in the long run, but either way, the effects will depend on *how* students use them. Consequently, we need to study different ways GenAI can impact how students interact with and learn chemistry.

Generative AI tools can potentially support chemistry students by adapting to their knowledge and assisting with lab reports,[5,6] problem-solving,[7−9] and programming in a disciplinary context. For instance, GenAI can support student self-efficacy and motivation in open-ended problem-solving,[10] promoting student agency. However, there is very little research studying the affordances of GenAI tools in chemical modeling.

In this study, this gap in the literature is addressed, exploring how a conversational generative AI affects students' modeling processes when solving computational chemistry problems. Modeling is important both as an educational strategy and as a scientific practice, and it is a complex process in which students need to draw on a multitude of skills, iteratively refining their ideas and testing them out.[11−15] This process fosters deeper conceptual understanding, systems thinking, and critical

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problem-solving skills, as students consider how different components interact within a system.[16−22]

Computational modeling, as a subfield of scientific modeling, involves making assumptions and simplifications, representing models mathematically, and designing code structures that implement these models.[23−26] Iterating on models as new insights emerge mirrors authentic scientific inquiry and promotes a more nuanced understanding of how chemistry works, providing insight into structure and dynamics at the molecular level.[27] Given its dual value as both a learning strategy and a core scientific skill, computational modeling is a powerful approach in chemistry education.[14,28−30] However, many students lack the computational background needed to fully engage in such activities. This study examines whether and how GenAI can help students overcome this barrier and engage more productively in the modeling process. Thus, the following research question is addressed: *Under what conditions does the use of generative AI based on LLMs support or hinder productive engagement with computational modeling in chemistry?*

Computational modeling in chemistry was chosen as the context for this study because it combines core disciplinary knowledge with programming and problem-solving practices. Computational chemistry is also increasingly recognized as a core component of modern chemical research and practice, and is consequently finding its way into chemistry education in both secondary and higher education.[27,31−34]

■ LITERATURE REVIEW

Generative AI dates back to the 1950s, but public interest surged after OpenAI released ChatGPT on 30 November 2022.[35] As users quickly discovered, recent advancements in algorithms and computing power have allowed companies to develop very versatile models, whose applications touch nearly every discipline.

While tools like ChatGPT are often referred to as conversational AIs because of their interactive nature, or as LLMs in technical contexts, we use the broader term *GenAI* in this article to emphasize their generative and pedagogical potential in educational settings.

The rapid development of GenAI has led to a surge of research debating opportunities and risks for education.[36−38] Some also provide specific examples or exploratory studies of GenAI behavior and problem-solving.[7,39,40] Several of these papers frame the generative AI as a student, studying how it performs on tasks related to specific scientific topics, like writing or engaging in problem-solving in chemistry and related scientific disciplines.[7,41−46] Beyond these discussions, studies report current classroom use and perceptions of GenAI;[38,47−52] students often use ChatGPT when stuck[49] and many apply it critically alongside other sources.[49]

Other researchers have begun empirically probing the affordances of GenAI for teaching and learning.[53,54] For instance, GenAI has the potential to support critical thinking skills in chemistry, and to improve learning by providing personalized and adaptive learning experiences.[4,41,55]

As this body of research makes clear, to critically evaluate the opportunities and risks of GenAI use in chemistry education, we must consider the affordances of the tool, that is, its capabilities and limitations. However, because generative AI tools' capabilities are highly context-dependent (that is, the performance of these tools is heavily based on prompting, context, and application)[56] it is also important to consider the conditions for *when* they perform poorly or well. For example, several researchers point out that GenAI tools may give students opportunities for interactive and adaptive learning,[36,37,57,58] while at the same time describing how a lack of contextual understanding and poor understanding of prompting-skills might hinder these processes.[59] This adaptive learning, as well as the caveats, extends into a computational context where GenAI could be used to help students understand, modify, and write code.[48,60,61]

Furthermore, while research on what GenAI can do is valuable in assessing possible uses of generative AI in chemistry, its relevance is often limited by a focus on the capabilities of specific GenAI tools at a specific point in time. Because capabilities change quickly, we focus less on what tools can do and more on strategies students use productively (or not) in chemical modeling. These strategies may be more resistant to obsolescence with future GenAI development.

■ THEORETICAL FRAMEWORKS

We adopt the Distributed Cognition (DC) framework to analyze student−GenAI interactions, and Weintrop's Computational Thinking taxonomy[62] to classify actions that students engage in.

Distributed Cognition

Distributed cognition (DC) is a theoretical framework from the learning sciences that posits that cognitive processes are not confined solely to an individual's mind, but are rather distributed across individuals, objects, and environments.[63,64] DC was developed by Edward Hutchins through his studies of navigation practices in planes and on naval vessels.[65,66] In his study of pilots, Hutchins frames the entire cockpit as a *cognitive system*, consisting of both equipment and the pilots themselves.[65]

In this study, the distributed cognition (DC) framework provides a useful lens to explore how students interact with GenAI as a cognitive partner. Conversational GenAI tools like ChatGPT add a new dimension to DC by dynamically responding to user input, not just supplying information but actively participating in thinking processes. This makes dialogic interaction with GenAI particularly relevant for complex activities like computational modeling, which requires the integration of chemistry, mathematics, and programming.

Distributed Cognition and Learning in Chemistry

DC has been applied in chemistry education, including studies of how students use laboratory equipment to build understanding.[67−70] However, distributing cognitive resources does not necessarily imply learning. Hutchins studied how pilots are more productive with the use of different artifacts and each other, not how pilots learn. Nonetheless, engaging productively in a distributed cognitive system has the *potential* for learning through creating scaffolds, using different representations, and engaging in collaborative activities.[68,71,72] Also, students should learn how to engage productively in a distributed cognitive system, for example with other people, laboratory equipment, and generative AI, because they are most certain to engage in such systems in their professional careers as chemists.

Belland[72] emphasizes the need for effective scaffolds for problem-based learning (PBL) to avoid confusion and resignation during complex problem-solving. Scaffolds can both simplify, activate students' pre-existing knowledge, and draw students' attention to important aspects of the problem.[72]

Belland further discusses the role of scaffolds as a part of a distributed cognitive system. He highlights the importance of the "agent controlling the executive function", which in the distributed system in this study would be the student. This agent cannot be removed without compromising the function of such a cognitive system.[72] Thus, students must retain the executive role; outsourcing it to the AI undermines learning.

In short, distributed cognition posits that knowledge emerges through interactions within a cognitive system and is spread across or embedded in the dynamic relationships among its components. This ultimately leads to the cognitive system achieving more than its constituent components. By applying the distributed cognition framework to the study of student interactions with generative AI, the aim is to uncover how this technology can act as a more knowledgeable other[73,74] to help − or hinder − students engage productively with computational modeling in chemistry.

Computational Thinking

While the primary analytical lens of this study is Distributed Cognition, the computational thinking (CT) taxonomy developed by Weintrop et al.[62] is used to support the analysis of student interactions with GenAI. CT, broadly defined, refers to the problem-solving practices associated with expressing processes in ways that a computer can execute.[75]

The Weintrop framework operationalizes CT in science education through four categories: data practices, modeling and simulation practices, computational problem-solving practices, and systems thinking practices. In this study, these categories were used deductively to classify the types of actions students engaged in when interacting with code and simulations during modeling tasks.

■ DATA COLLECTION AND ANALYSIS

Data Collection

In line with the goals of studying the dynamics between students and generative AI, a cognitive clinical interview methodology[76] with a think-aloud protocol was used, in which pairs of students solved a multipart computational chemistry problem with the aid of a conversational generative AI tool (OpenAI's ChatGPT-4.0), and the interviewer present. Clinical interviews are useful for analyzing how meaning is negotiated and constructed, both individually, between individuals, and between individuals and external tools, and have a long history in studies of problem-solving.[77,78] Pair-interviews of students were conducted to increase the odds of students discussing with each other and explaining how they think.

The study took place at the University of Oslo (UiO), Norway in the winter of 2023/2024. At UiO, all first-year STEM students learn programming, and most bachelor-level physics/chemistry courses integrate it as a disciplinary tool.[79]

Students were recruited randomly for this study through an e-mail to all chemistry and physics students at the University of Oslo. Twenty-one students (10 females, 11 males) in total participated in the interviews, of which 8 were chemistry students (including 1 student of material science) and 13 were physics students, all in the second or third year of their 3-year bachelor's degrees. While the goal was to recruit a balanced sample from both disciplines to study disciplinary differences, the chemistry program is considerably smaller than the physics program at the University of Oslo. Despite contacting all students in both departments, only eight chemistry students volunteered, so the sample was supplemented with additional physics students. This was a pragmatic decision based on participant availability rather than a theoretical choice. This imbalance has been considered in the analysis and reflections on disciplinary differences have been included, particularly in relation to prior knowledge, where relevant.

Although students were recruited in pairs, three students participated individually due to groupmates not showing up. The interviews lasted approximately 1.5 h each. During the interviews, the interviewer sat on the opposite side of the students, prompting reflection and discussion, but interfering as little as possible with the problem itself. Whenever possible, the interviewer prompted the students to use GenAI to help themselves. The interviewer used a semistructured approach without a fixed interview guide. Prompts were given spontaneously to support reflection, not to guide specific actions. Typical prompts included "Why do you think that?" or "Could the AI help here?". While this allowed for a natural flow, variation in wording may have influenced students' responses. To reduce bias, the same interviewer conducted all sessions and kept prompting style consistent.

The students had access to a computer with a programming environment and a browser with OpenAI's ChatGPT-4.0. No personalized pre-prompts or prompting strategies were included for the students to maintain a more naturalistic context.

In the first part of the interview, the students were asked to describe their use of generative AI in their courses. This was done to make it possible to compare self-reported use with actual use in an experimental setting.

In the second part of the interview, the students were asked to solve three computational problems at the boundary between chemistry and physics, which focused on the skills of interpreting, extending, and constructing computational models.[80] The three problems were designed by the first author to assess student computational thinking with different degrees of scaffolding. The first problem was designed as an interpretation task, the second as an extension task, and the third as a construction task. Due to the scope of this study, the results from the construction task were omitted from this paper.

The task was designed as an open-ended task with minimal scaffolding provided. First, the students were supposed to interpret a complete MD simulation of argon gas. Then, they were asked to modify the simulation to account for O_2 instead of argon. Last, they were asked to make a simulation that shows how the concentration of a chemical reaction varies with time. An example snippet of the code is shown in Figure 1. The complete code and task are included in the Supporting Information.

```
def generate_velocities(N, T, m, kb=1.38064852E-23):
    v = np.random.randn(N,3)*np.sqrt((kb*T)/m)
    P = np.sum(m*v,axis=0)
    v = v - P/(m*N)
    return v
```

Figure 1. Example of a function used in the simulation.

All interviews were video recorded, transcribed, and anonymized. Although the interviews took place in Norwegian, excerpts used in this article have been translated into English. In addition to interview recordings, chat logs generated by the students with ChatGPT were also collected. The project was approved by the Norwegian Agency for Shared Services in

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Table 1. Coding Scheme Used in the Analysis

Code	Description
Productiveness	
Productive	Use of GenAI where the output is in alignment with the purpose of the task. The output provides a valuable piece of information for the problem-solving process that is used in conjunction with the students' own cognitive resources. Students retain the executive function.
Unproductive	Use of GenAI where the output is not in alignment with the purpose of the task or hinders the students in using their own cognitive resources. The output can also be incompatible with students' previous knowledge or lead to cognitive overload. Students transfer the executive function to the GenAI.
Actions	
Visualizations	Students interpret or create visualizations.
Modeling	Students construct, evaluate, or extend models and simulation output.
Computational Problem-Solving	Students engage in using and evaluating suitable programming strategies, troubleshooting and debugging.
Systems Thinking	Students move between levels of the system: macroscopic, submicroscopic, and symbolic.

Education and Research (SIKT), which regulates GDPR and ethics in research in Norway.

Analysis

Analysis of the interviews took place in three phases. During the first phase, the first author deductively coded the interview transcripts using an analytic framework that operationalized the distributed cognition theoretical framework. This analysis was triangulated with chat logs to distinguish between the knowledge contributed by the students and the insights provided by ChatGPT.

The coding frame used for this first round of analysis is presented in Table 1. This framework was developed a priori, based on the central constructs of the Distributed Cognition framework (e.g., executive function and human-tool interaction). The codes "Productive" and "Unproductive" reflect these important aspects of the main framework.

Some of the codes were also constructed to describe the relevant actions that students engaged in. A slightly modified version of the four categories of Weintrop et al.'s computational thinking taxonomy (2016) was used to classify student actions during modeling. A deductive approach was chosen to ensure that the analysis aligned closely with the theoretical framework, enabling a systematical identification of how different cognitive roles and functions were distributed between students and GenAI. This approach was especially suitable given the goal of distinguishing productive and unproductive configurations of the distributed cognitive system, as defined by prior theory.

More specifically, episodes were identified in which GenAI contributed to the cognitive system, either productively or unproductively. Episodes where students were actively engaged in the modeling process using their own cognitive resources in combination with the output from the GenAI while retaining the executive function, were classified as productive. For example, some students recognized output from the GenAI as a concept they knew from chemistry, which in turn helped them connect their thinking about the problem to the GenAI output.

On the other hand, episodes where students made clear that they either did not understand the output from the GenAI or where they solely relied on GenAI for the solution were classified as unproductive. In such situations, the students lost control of their executive function to the GenAI and were hindered from using the information from the GenAI in the modeling process.

In the second phase, the first author performed a second round of analysis to identify specific actions the students were using to solve the problems. This analysis focused on the different actions students took when solving the modeling tasks, which was done to strengthen the claims of which episodes were productive and which were unproductive.

Descriptions of the codes are provided in Table 1. An episode spans from one GenAI output to the next and can contain none or several actions. The actions were coded as mutually exclusive.

Before proceeding with the analysis, the code categories were discussed with the second author, and the second author coded approximately 20% of the material, randomly sampled from the different interviews. This resulted in a Cohen's Kappa of 0.70, which signifies a moderate agreement.[81] The disagreements were mostly related to computational problem-solving vs modeling, and the criterion for this code was consequently discussed and specified.

The coding frame was subsequently revised to distinguish more clearly between these two categories: *computational problem-solving* was defined as programming-specific skills (e.g., syntax, debugging), while *modeling* involved evaluating or constructing representations of chemical systems. This refinement addressed the source of disagreement, and it was judged that inter-rater alignment on the remaining categories was sufficient for the purpose of this study.

In the third phase, the findings from the previous steps were integrated by conducting a thematic analysis.[82] This analysis aimed to uncover recurrent themes related to what characterized the productive and unproductive episodes. This was achieved by identifying distinctive patterns that defined productive and unproductive interactions, based on how the students contributed to the cognitive system. This culminated in the identification of four distinct themes. Two themes cover productive use of GenAI:

- Theme 1: Leveraging AI to retrieve and clarify information
- Theme 2: Using GenAI to test and critique ideas

The remaining two themes were circumstances under which generative AI use was unproductive:

- Theme 3: Outsourcing thinking to the AI
- Theme 4: GenAI output exceeds student understanding

In what follows, each of these patterns is described, illustrating them with examples from the data set.

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RESULTS

Productive Uses of GenAI in Chemical Modeling

Both productive strategies featured a common element: students used the GenAI in a purposeful manner that was aligned with their own cognitive capabilities. In other words, prior to interacting with the GenAI agent, students first familiarized themselves with the code and problem they were trying to solve. For example, students spent time trying to comprehend the code, understand how it represented the system, or formulate a hypothesis about how the system worked based on what they already knew or what they could infer from the code. Building on this understanding, students then leveraged the GenAI to fill in specific gaps in their knowledge or to interpret the parts of the code that they had identified as difficult to comprehend.

Students using GenAI productively did gradually build a better understanding of the connections between the code, the mathematical model, and the chemical system; that is to say, in these productive examples, students use GenAI to supplement their understanding rather than asking the GenAI to provide complete answers or solutions.

Below, these two themes are described in detail, illustrated with excerpts from the data set.

Theme 1: Leveraging AI to Retrieve and Clarify Information. In many productive instances, students used GenAI to retrieve and reinterpret information that meaningfully connected with what they already knew. These students were able to integrate the AI's output into a larger cognitive system because they possessed knowledge of relevant disciplinary concepts and models. Drawing on prior knowledge allowed them to evaluate the relevance of the AI output, ask targeted questions, and make sense of technical representations, such as unfamiliar equations or force functions. From a distributed cognition perspective, GenAI functioned as an external memory or interpretive scaffold, but the coherence of the cognitive system hinged on the student's ability to recognize and build upon the output. In this way, GenAI served not as a surrogate thinker, but as a cognitive amplifier, helping students elaborate on concepts, visualize systems, and connect microscopic models to symbolic code.

Andrew, a materials science student, exemplifies this theme. In the initial part of his interview, he explicitly talked about how going back and forth in a dynamic discussion with the AI, or with other students, is his primary choice of studying. In the second part of the interview, he uses this strategy extensively when working on the exercises: using his own background knowledge as a foundation and building upon this with bits and pieces of information from the AI. After coming to a general conclusion on what the code does, he starts digging into the force function, which many students struggled with in the first exercise. He asks the GenAI to interpret what force model is used in the calculation of forces by supplying the expression of the force model used in the simulation:

> ***Andrew:*** *I don't know if it makes sense to put in [to ChatGPT] the whole force equation and ask 'what force is this?' [prompt: "24*epsilon*(2*sigma**12/rdist**13 - sigma**6/rdist**7)*dr/rdist what force is this"]*
> *[reads] It suggests 'Lennard-Jones potential' (...) Yes, I've heard of that. We are talking about intermolecular dynamic simulations. Epsilon is the depth of the potential well.*

Here we see that the GenAI output activates Andrew's pre-existing knowledge of the Lennard-Jones potential. This, in turn, guides his actions further to visualize how the LJ potential varies with intermolecular distance, and to engage further in a discussion about what forces underlie this potential, engaging in modeling and systems thinking.

The interaction between Andrew and the AI, including the relevant actions performed by Andrew (following the coding scheme from Table 1), is summarized in Figure 2, and excerpts from the interview are provided as SI.

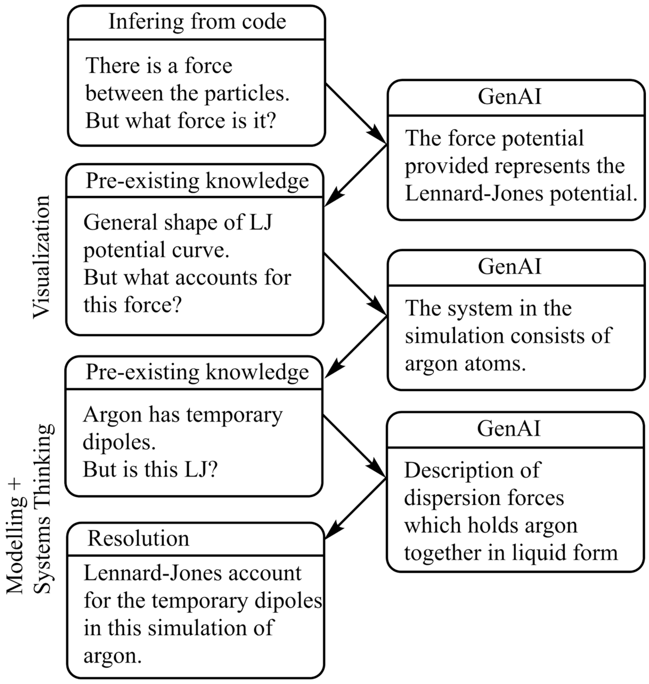

Figure 2. Workflow of the student Andrew working on interpreting code, dynamically using the GenAI to provide information which led him to engage in productive actions.

This example illustrates how students can use GenAI tools as part of a distributed cognitive system, helping them retrieve and reinterpret information in ways that connect their internal knowledge with external knowledge provided by the AI. In this system, GenAI acts as a dynamic external resource that students can integrate into their thinking process. When used productively, such tools extend the cognitive resources available to the learner for solving a problem. This configuration is illustrated conceptually in Figure 3.

Theme 2: Using GenAI to Test and Critique Ideas. A second theme illustrates another aspect of productive use. Rather than merely relying on prior knowledge to make sense of AI output, some students actively managed the AI, using it as a tool to clarify and test ideas by refining prompts, steering responses, and critically comparing outputs. In doing so, they framed GenAI not just as an information source, but as a tool to be managed and orchestrated, retaining their executive control.

Carl provides a clear example of a student who retains the executive function while engaging in dialogue with GenAI. He begins by asking ChatGPT what the overall code does, then uses this overview to steer his own thinking. Rather than accepting the answer at face value, he uses it to guide his next steps:

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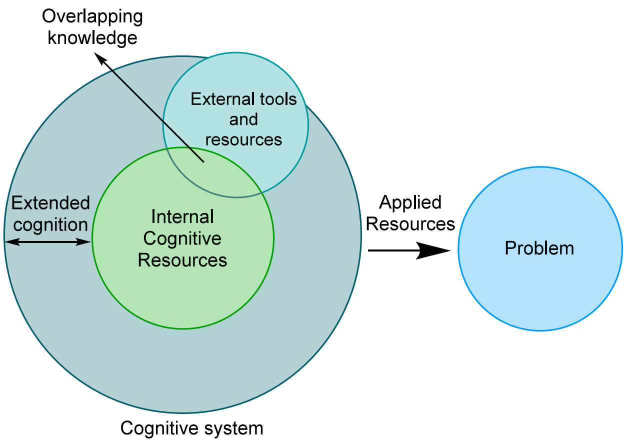

Figure 3. Conceptual representation of a distributed cognitive system. Internal cognitive resources (e.g., prior knowledge, reasoning strategies) can be expanded through interaction with external tools and resources (e.g., GenAI). This interaction is productive when there is a considerable overlap between the knowledge provided by the AI and the knowledge of the students. When students actively coordinate these elements, the extended cognitive system enables deeper engagement with tasks such as modeling. The arrow indicates how the extended cognitive system contributes to productively solving the problem.

> ***Carl:*** *[studies the code after skimming the explanation from the GenAI] Okay, so this [points at the force equation in the code] is a force based on the Lennard-Jones potential. [...] So, we're basing it on the Lennard-Jones potential when we calculate these forces, and we have these 'attractive forces' and 'repelling forces' [points to the code]. And then we calculate them for each point in space, because here [points to the code] we generate positions. Then we integrate them.*

Carl is first familiarizing himself with the function that calculates the forces between particles. He uses this overview to connect the different explanations provided by the GenAI to each line of code. This also prompts him to run the code, getting a visual evaluation of the code output. Based on this information, Carl reads the code more closely and summarizes what the code does:

> ***Carl:*** *It seems as if we have some kind of particle simulation that shows the movement of particles over time, according to molecular dynamics and especially the Lennard-Jones effect. It's just a hypothesis. I haven't read the code that closely yet. We have random starting conditions that we simulate; we just initiate particle positions and velocities, and it's, as I said, random. And then we simulate the forces that act between the particles, on each other. That defines the Lennard-Jones forces [information provided by ChatGPT]. And then we integrate the forces over time, I think, and then we plot it. That's my hypothesis on what the code does. But I hardly read the summary···*

Carl then proceeds to zoom in on specific functions in the code and cross-references these with the GenAI output, using it not as an authority but as a sounding board to refine his interpretation. He moves between code, mathematics, conceptual explanations, and graphical output − coordinating different representations of the same model. He is also maintaining control over the system's cognitive direction by not transferring the executive role to the AI.

Lucas and June's approach similarly exemplifies the student-orchestrated use of GenAI. When interpreting the force function, Lucas hypothesizes that the model simulates the Lennard-Jones potential and turns to ChatGPT for confirmation:

> ***Lucas:*** *[reading the code] Two sigma raised to the power of 12 divided by... I mean, it really reminds me of Lennard-Jones potential. [...]*
> ***Interviewer:*** *What's the Lennard-Jones potential?*
> ***Lucas:*** *It's the potential between molecules... I don't remember exactly, but it's like the characteristic 'R' raised to the power of 6 minus 'R' to the power of 12, or the other way around, and it gives attraction at distance, but it gives a repulsion if they're very close together so that they don't keep... so that it gives an appropriate characteristic distance between the molecules, were they to be completely stable. [...]*
> ***Lucas:*** *[Asks the AI what the force function does. Skims and reads] 'The code snippet defines a function for a system called... [exclaims] 'Lennard-Jones! Yay!*
> ***June:*** *Let's go!*

After receiving confirmation from the GenAI, the students celebrate, indicating that the tool is not providing *new* information, but rather validating their own inference. This suggests that GenAI is being used to support metacognitive processes − evaluating confidence in a hypothesis, reinforcing conceptual links, and allowing students to continue with a clearer mental model. This type of dialogic use aligns with the DC view of cognitive scaffolding, where the tool helps stabilize emerging ideas, without taking over the reasoning process.

Across these examples, students are not passive recipients of information but active participants in a dialogue with the GenAI tool. They use it to test hypotheses, clarify confusion, or confirm that their interpretations are on the right track. This contrasts with unproductive use, where students defer judgment to the AI or use it prematurely. In the productive episodes, the GenAI acts more like a collaborator than a tutor − an external voice in a dialogic process that remains under student direction. This orchestrated, back-and-forth interaction exemplifies a well-regulated distributed cognitive system, where tools are used flexibly and purposefully, in service of student-led inquiry.

Unproductive Uses of GenAI in Modeling

While many students used GenAI in ways that supported them in the modeling process, clear patterns of unproductive use were also identified. In these cases, interactions with the GenAI did not lead to deeper engagement with the modeling task, nor did they support meaningful progress toward a solution. Instead, students either relinquished control of the problem-solving process to the AI or received output that exceeded their capacity to interpret and apply. In what follows, two themes that characterize these breakdowns are presented.

Theme 3: Outsourcing Thinking to the AI. While the previous two themes illustrate how GenAI can be purposefully integrated into a distributed cognitive system, not all student interactions followed this pattern. In several interviews, the opposite dynamic was observed: students handed over control of the problem-solving process to the AI, relying on it not as a scaffold but as a surrogate problem-solver. In these cases, the generative AI was no longer a component within a student-led system, but rather the system's de facto driver. This transfer of executive control from the student to the GenAI marked a breakdown in the distributed system's integrity, resulting in unproductive interactions.

For example, Tom and Ben initially inferred some basic information about the code but encountered difficulties when trying to describe the forces between particles. They struggled to move between the symbolic level of the code and the microscopic level of molecular interactions, an essential feature of systems thinking. Seeking help, Ben turned to the GenAI:

> ***Ben***: *[Prompting after copying the whole code into ChatGPT: "What causes the particles' behavior?"] Yes, so it lists a whole bunch of forces that affect the behavior of the particles here.*

Although Ben intended to ask about interparticle forces, the GenAI interpreted the question broadly, producing a comprehensive explanation of all components in the simulation − from initialization to integration and visualization. The output overwhelmed the students, rather than helping them. As physics students with limited background in intermolecular forces, they lacked the disciplinary context to filter the output or judge its relevance. The GenAI response thus remained unintegrated into the cognitive system − informationally rich but not productive for engaging in modeling and learning.

This pattern was common in Task 2 (the extension task), where students were asked to modify the model to simulate oxygen molecules instead of argon atoms. In contrast to Task 1, where most students used GenAI after some analysis or interpretation of the problem, many students prompted the GenAI immediately in Task 2 − without attempting to scope the system or identify relevant variables themselves. In effect, they offloaded the abstraction process to the AI, asking it to both define and solve the problem. In doing so, they surrendered the executive role within the cognitive system, allowing the AI to determine not just *what* information to provide but also *how* the problem should be framed.

For instance, Isaac and Eve began task 2 with the broad prompt "change argon to oxygen what happens?", which resulted in a wide-ranging and mostly irrelevant answer. While this is also an example of cognitive misalignment (discussed in theme 4), it simultaneously illustrates a breakdown in agency: the students allowed the AI to set the scope of their modeling task before they had made sense of the problem themselves. As a result, the GenAI introduced complexity without direction, and the students' interaction with the tool became reactive rather than strategic.

On top of the student difficulties in parsing the GenAI output, imprecise prompts tended to result in responses with errors. For example, in addition to suggesting that students update their model to include paramagnetism, vibration, rotation and "other [undefined] complex terms", the GenAI suggested to a pair of physics students (Kyle and Greg) that they should add models for polar molecules, even though O_2 is nonpolar. Their physics background made them unable to evaluate this information critically, leading them into confusion and further unproductive uses of the AI.

From a distributed cognition perspective, these episodes illustrate what happens when students stop regulating the flow of information within the system. Rather than retaining their executive functions, the students defer responsibility for problem-solving to the GenAI. When this happens, the GenAI is no longer a scaffold that extends student cognition; it becomes the agent driving the system. The cognitive load on the student decreases, but so too does the opportunity for learning, reflection, and deeper engagement.

Theme 4: When GenAI Output Exceeds Student Understanding. Another common reason for unproductive interactions with GenAI was a mismatch between the information provided by the tool and the students' level of understanding − *a cognitive misalignment*. In such cases, the generative AI produced content that students could not interpret or connect with their existing knowledge, resulting in cognitive overload or confusion. This misalignment often stemmed from prompts that were too vague or overly broad, which failed to constrain the AI's meaning space to a manageable and relevant scope.

For example, as briefly mentioned in the discussion of Theme 3, Isaac and Eve's general prompt led ChatGPT to explain not only physical differences between argon and oxygen but also broad topics such as environmental impact and chemical reactivity. Rather than helping the students narrow their focus, the GenAI introduced additional complexity into their cognitive system:

> ***Isaac:*** *But [infers from the AI output] the main point is 'chemical reactivity'.*
>
> ***Eve:*** *Yeah, okay, but that answers...does that answer our question?*
>
> ***Isaac:*** *Partially? It simply just says that we must take into account that oxygen can and will bind together with everything possible [...]*

The GenAI's response, while informative, did not align with the learning goals or the specific context of the task. Instead of supporting modeling by aiding them in abstraction, it introduced chemistry topics like reactivity and bonding that were beyond the intended modeling scope. Modeling the breaking of chemical bonds requires quantum chemical potentials, which are very advanced to implement as computer code. In DC terms, the tool's contribution could not be meaningfully integrated into the system because it exceeded the boundaries of the students' existing knowledge and goals.

From a DC perspective, this theme illustrates that productive use of external tools like GenAI depends on how well the tool's output connects back to what students already know, and how clearly the task is defined. When students lack the necessary background knowledge to understand the output or when they have not clarified the goals or limits of the problem − the system becomes overloaded with information that is difficult to interpret or use. In such cases, GenAI no longer functions as a helpful scaffold, because the information it provides cannot be meaningfully taken up or applied by the students.

Distribution of Categories

In total, 141 instances of student-GenAI interaction were identified across all sessions. Of these, 95 interactions (67.4%) were coded as *productive*. The remaining 46 instances (32.6%) were coded as un*productive*, such as off-topic prompts, overly general queries, or overreliance on GenAI without reflective follow-up. While most students predominantly engaged in productive uses, *productive and unproductive strategies often co-occurred within the same session*. This suggests that effective use of GenAI was not a fixed trait of a user, but rather a fluctuating pattern influenced by task complexity, prior knowledge, and prompting strategy.

Sequential inspection of the coded transcripts also revealed that productive uses of GenAI were often immediately followed by instances of computational thinking (CT) practices. In contrast, unproductive uses were frequently

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terminal, that is, not followed by further engagement or progress in the modeling task. This pattern indicates that productive prompts often precede or lead into more sophisticated modeling and reasoning practices.

When disaggregating the data by discipline, a noteworthy pattern emerged: chemistry students engaged more productively with GenAI than physics students. The frequencies across all interviews are shown in Table 2. One plausible explanation is the difference in domain-specific knowledge. Since the modeling task was grounded in chemical phenomena, chemistry students were better positioned to recognize relevant terminology, evaluate GenAI outputs, and build meaningfully on them. In contrast, several physics students struggled to connect the AI-generated information with appropriate disciplinary concepts, which may have limited their ability to use the tool productively.

Table 2. Frequencies of Productive and Unproductive Uses of GenAI across All Interviews

	Chemistry students		Physics students	
	Frequency	Percentage	Frequency	Percentage
Productive	52	86.7	43	53.1
Unproductive	8	13.3	38	46.9
TOTAL	60	100	81	100

■ DISCUSSION AND IMPLICATIONS

When used productively, GenAI can scaffold complex, iterative computational modeling. This scaffolding could provide students with the opportunity to engage in authentic modeling tasks that prepare them to think as scientists.[26]

This study identified four student-GenAI interaction patterns. In two productive themes, students either used GenAI to retrieve and clarify information, drawing on chemistry knowledge to interpret and connect AI output to chemical concepts, or to test and critique ideas, using metacognitive strategies to iteratively prompt and guide the AI as a reasoning partner. These themes show the importance of domain-specific content knowledge and strategic use of AI to engage productively in modeling with GenAI.

In contrast, some students used GenAI in ways that disrupted rather than supported their thinking. In some cases, they ceded control to the AI, relying on it to define the task, interpret the problem, or generate answers without reflection or evaluation. In other cases, they received responses that were too complex or tangential to understand or apply, often because they turned to GenAI before fully framing the problem themselves. These patterns led to shallow engagement and stalled progress in the modeling process.

How Can We Use GenAI Productively in Chemistry Education?

As the ongoing discussion about GenAI in chemistry education demonstrates, educators must be aware of both the affordances and pitfalls of using GenAI for learning and problem-solving. Because many students already use GenAI,[83,84] guidance should emphasize using GenAI to support (not replace) reasoning and learning.

Students could thus benefit from guidance on how to use AI productively. As shown in this study, relying too much on copy-pasting questions instead of breaking the problem down into smaller chunks, leads to unproductive interactions. This problem has been shown to occur in other contexts as well,[59] and students consequently need guidance in how to productively work with, and in particular learn, with generative AI. Based on the findings in this study, the following set of guidelines is suggested:

1. **Think first, ask later, then think again.** Students who engage in initial abstraction and analysis before turning to GenAI were more successful in using its output because the output fits better with their own understanding.
2. **Give GenAI enough appropriate context to limit its room for interpretation.** The more specific the prompt is, and the more grounded it is in students' own knowledge, the more relevant and useful the GenAI output will be. Vague and general prompts often lead to less relevant output, hallucinations, or unnecessarily complex suggestions.
3. **Interact dynamically with the GenAI.** Productive interactions were not one-off prompts but iterative conversations.

The kind of iterative engagement and retrieval of knowledge outlined above are well-known strategies for learning.[85,86] The dialogic nature of GenAI – its ability to generate different responses in different contexts – makes it uniquely suited for supporting such back-and-forth interactions. Unlike static resources, GenAI can respond in real-time to students' prompts, restate information at different levels of complexity, or adopt different "voices" (as a peer, tutor, expert, etc.).[87] These features allow GenAI to serve as a flexible tool for self-scaffolding, helping students build bridges between what they already know and what they are trying to understand. However, students need guidance on when and why to use GenAI. Our findings identify productive patterns and conditions that can inform such guidance.

The results from our study may also be interpreted through the lens of cognitive load theory, both seen from an individual's perspective[88,89] and collaboratively,[90] possibly investigating how different types of cognitive load[91] affect learning. A more detailed discussion of these findings through the lens of cognitive load theory is provided in the Supporting Information.

One important direction for future implementation involves the use of customized AI assistants, which allow users or instructors to define the role and behavior of the GenAI through a persistent pre-prompt. The results from this study suggest that defining the AI as a *learning assistant* – for example, prompting it to ask guiding questions and help students retrieve knowledge rather than provide answers – could help students remain in control of the cognitive process. For instance, students could use the recently introduced "study-mode" in GPT 5, which is framing the GenAI as a scaffold for reflection and problem-solving rather than a shortcut to answers. Educators are encouraged to explore these affordances to help students engage more productively with GenAI tools.

■ CONCLUSION

By identifying the four themes in this study, a conceptual vocabulary and empirical basis is proposed for chemistry education practice and future research into generative AI in chemistry education, possibly extending into science education in general. These themes suggest multiple promising avenues for further exploration.

First, future research should investigate how students can be supported in retaining executive control when using GenAI. The findings from this study indicate that productive use depends not just on the capabilities of the AI tool, but on students' capacity to direct it. This raises important questions about what kinds of instructional scaffolds, training interventions, or interface designs best support students in orchestrating such systems.

Second, while this study described instances where students used GenAI as a scaffold for interpretation and model-building, it remains an open question whether and how these interactions contribute to long-term learning. Does dialogic interaction with GenAI promote conceptual change or durable understanding, or does it merely aid short-term problem-solving?

Third, the results from this study highlight the importance of students' pre-existing disciplinary knowledge in shaping their interactions with GenAI. When students lacked a foundation to interpret GenAI output, the system broke down. This suggests a need for research on how domain knowledge and AI literacy codevelop, and how they might be jointly scaffolded. For example, can instructional sequences be designed to progressively build both disciplinary and AI-interaction competencies, so that students become more adept at using GenAI as their understanding deepens?

Finally, future research could attend more closely to *breakdowns* in distributed cognitive systems. Much of the existing literature, including Hutchins' original studies, focus on how a distributed system aids individuals in performing tasks, or how it supports learning under ideal conditions. But as shown in this study, breakdowns in student-AI interaction are quite common, especially when faced with complex open-ended tasks. Studying these breakdowns systematically, across contexts, disciplines, and student populations, may help us refine theoretical models of distributed cognition and inform more robust guidelines for educational AI design and classroom integration.

Limitations

Due to the exploratory nature of this study, the results are not meant to give a complete overview of the different mechanisms inherent to student thinking and GenAI interaction. Rather, the aim was to capture general patterns in student interaction with a conversational generative AI.

Since both chemistry and physics students are included in the sample of this study, the sample size is quite small with regards to the chemistry students. This might limit the insight we gain into how different chemistry students use AI to do the modeling task, while at the same time providing information about the importance of domain-specific knowledge.

The chosen task and research design also provide limitations to these results. The task was deliberately designed to be challenging, with each part increasing the level of challenge. When faced with complex problems in the context of this study, most students tended to use AI prematurely, especially in tasks 2 and 3 where the complexity was highest. Because of this, these results are not necessarily representative of how students will typically use GenAI in every aspect of chemical modeling. Furthermore, the experimental conditions were subject to a very short time limit (1.5 h), which certainly affects student use of GenAI. At the same time, this restricted setting might provide valuable insights that may still apply outside of an interview space: when faced with complex exercises and time pressure, students may tend to use GenAI unproductively. This is important for chemistry educators to recognize, especially if they are considering giving students access to GenAI tools during time-limited assessments.

It is also important to address the interviewer's role in the distributed cognitive system. In some parts of the interview, the interviewer had to intervene to ask questions or alleviate the rising conflicts between the output from the GenAI and the students' knowledge. As prompts were given spontaneously rather than from a fixed guide, variations in wording or timing may also have influenced student behavior. While efforts were made to keep prompting consistent, this remains a limitation. More research will be needed to examine the use of GenAI in other settings in chemistry, but the results of this study can provide valuable guidance on what to study in such contexts.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available at https://pubs.acs.org/doi/10.1021/acs.jchemed.5c00657.

Additional Results and Analysis (PDF, DOCX)

Interview Protocol with Complete Task (PDF, DOCX)

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Notes

Generative AI tools (OpenAI's ChatGPT) were used to support the writing process of this article in two specific ways: (1) translating draft sections and interview excerpts from Norwegian to English, and (2) refining the clarity, structure, and coherence of arguments in selected parts of the manuscript. All content was critically reviewed, revised, and approved by the human authors to ensure accuracy and alignment with the intended meaning. AI was not used for the analysis or for handling data.

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